

# Chuong Dang TA

📍 Stockholm, Sweden | Willing to relocate

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## About Me

Energy systems and data science enthusiast with a thermal engineering background. Expected to graduate in **September 2026** from an Erasmus Mundus MSc (**DENSYS**). For my [Master's Thesis](#) at [rebase.energy](#) (Stockholm), I forecasted wind turbine icing power losses using an extended **IEA Wind Task 19** framework and **LightGBM**. Prior experience spans hybrid renewable system modelling and techno-economic policy analysis in Vietnam. Active builder of energy data side projects, including probabilistic day-ahead price forecasting, smart heat pump control, and energy storage optimization. Published author in 2 peer-reviewed Q1 journals (IFs: 11.8, 9.2).

## Education and Training

**University of Lorraine, France & Politecnico di Torino, Italy** - [densys.univ-lorraine.fr](https://densys.univ-lorraine.fr)

Sep 2024 – Sep 2026

**Master of Science in Decentralised and Smart Energy Systems (DENSYS)**

A two-year Erasmus Mundus Joint Master's program (120 ECTS) focusing on the modeling, control, and optimization of decentralised and renewable energy systems.

### Selected Coursework:

- University of Lorraine: [Data and Forecasting in Microgrids], [Optimal Local Design Energy Network].
- Politecnico di Torino: [Polygeneration and Advanced Energy Systems], [Smart Electricity Systems].

### Key Projects:

- Machine Learning (ML) Enhanced LCA of a North Sea Offshore Wind Farm*: Developed integrated LCA-ML framework achieving 12,600× computational acceleration ( $R^2=98.5\%$ ) for rapid environmental optimization across 285 design scenarios.

**Hanoi University of Science and Technology (HUST), Vietnam** - [hust.edu.vn/en](https://hust.edu.vn/en)

Aug 2018 – Jan 2023

**Bachelor in Thermal Engineering**

Graduated **1<sup>st</sup>/218** with a CGPA of 3.54/4.00 from Vietnam's leading technical university. Core studies covered thermodynamics, heat transfer, district heating and cooling systems, and power plant systems.

**Bachelor Thesis**: “*Techno-Economic Analysis of Internal Combustion Engines Using Diesel and LNG*” (Grade: 9.5/10)

## Scientific Research - ORCID: 0009-0009-7530-7430

- Dang-Chuong Ta; Thanh-Hoang Le; Long Van Phan; Hoang-Luong Pham, “Feasibility Analysis of Hydrogen Co-Firing in Vietnam's Gas Power Plants for the Period 2035–2050,” *Energy Conversion and Management*, Impact Factor: 11.8, **2025**. DOI: [10.1016/j.enconman.2025.120192](https://doi.org/10.1016/j.enconman.2025.120192)
- Dang-Chuong Ta; Thanh-Hoang Le; Hoang-Luong Pham, “An Assessment of the Potential for Large-Scale Hydrogen Export from Vietnam to Asian Countries: Techno-Economic Analysis, Transport Options, and Energy Carrier Comparison,” *International Journal of Hydrogen Energy*, Impact Factor: 9.2, **2024**. DOI: [10.1016/j.ijhydene.2024.04.033](https://doi.org/10.1016/j.ijhydene.2024.04.033)
- (Ongoing) Dang-Chuong Ta et al., “Wind Farm-Scale Forecasting of Turbine Icing Power Losses Using Multi-Ridge Regression Models,” based on MSc thesis research at [rebase.energy](#) - extending the turbine-level classifier-regressor framework with multi-ridge regression models and upscaling to wind farm-level forecasts.

## Work Experience

[rebase.energy](#), Stockholm, Sweden

Jul 2026 – Present

**Junior Data Scientist - Wave Energy Converter Power Forecasting**

- WEC Power Forecasting**: Autoregressive power forecasting via a **Multi-Ridge Regression** stack (**Elastic Net** and **LightGBM**).
- Wave Energy Flux Forecasting**: Forecasting wave energy flux from **ERA5 Ocean Wave** data, benchmarking regularised and tree-based models (**Elastic Net**, **Random Forest**, **LightGBM**) against probabilistic methods (**Quantile Regression**, **Quantile Regression Forests**, **Bootstrapped residuals**).
- Skills Learned**: Multi-Ridge Regression stacking, probabilistic forecasting, and working with ERA5 ocean wave reanalysis data.

[rebase.energy](#), Stockholm, Sweden

Feb 2026 – Jul 2026

**Master Thesis Student** - [Thesis Project Link](#)

- Icing Detection & Feature Engineering**: Labeled icing events on multi-year Swedish turbine SCADA data via an extended **IEA Wind Task 19 Sigmoid Performance Ratio** framework, corrected SCADA-ERA5-Land biases with **Quantile Mapping**, and used **Spearman/SHAP** screening to select stable predictors.
- Icing Power-Loss Forecasting**: Built a two-stage **LightGBM** classifier-regressor with walk-forward cross-validation to forecast icing losses at **1-36 hour** lead times, showing SCADA signals dominate near-term skill while NWP forecasts matter more for longer horizons.
- Skills Learned**: Time series forecasting under non-stationarity, and bias correction of NWP data against ground-truth SCADA.

### Junior Researcher

- **Hydrogen Integration Study:** Contributed to Vietnam's national hydrogen roadmap by conducting feasibility studies on the integration of green hydrogen into thermal power plants.
- **Techno-Economic Analysis:** Handled techno-economic modeling, hydrogen cost benchmarking, and policy analysis.

### Vietnam Initiative for Energy Transition (VIETSE), Hanoi, Vietnam

May 2023 – Sep 2023

#### Research Intern

- **Microgrid Modelling:** Modeled hybrid renewable systems (solar, wind, BESS) using HOMER Pro and Python.
- **System Optimization:** Analyzed system performance under different policy scenarios and contributed to policy briefs for Thai and Vietnamese stakeholders.

### Selected Projects - [chuongta.github.io/EnergyForecasting](https://chuongta.github.io/EnergyForecasting)

- **LLM Assistant for Operational BESS Dispatch**  
Integrated LightGBM day-ahead price forecasting with a Pyomo scheduling model, using a local LLM agent to query and explain optimal battery dispatch schedules.
- **Probabilistic Electricity Price Forecasting & Stochastic Storage**  
Developed DK1 day-ahead price forecasting models (Quantile Regression, Quantile Regression Forests, bootstrapped residuals) to drive a Pyomo stochastic battery dispatch model.
- **Smart Heat Pump Control**  
Built a physical model and Gradient Boosting digital twin of a heat pump, using Reinforcement Learning (PPO) to optimize thermal storage dispatch for price arbitrage.
- **Structure of Electricity Markets & EU/Nordic Market Design**  
Personal blog series covering European market design, clearing mechanisms, and deregulation of the Nordic electricity market.

### Technical Skills

- **Machine Learning:** Supervised, unsupervised, and reinforcement learning; hybrid, physics-informed models for energy system forecasting and predictive maintenance.
- **Data Science & Optimisation:** Applied MILP, MINLP, and heuristic algorithms (e.g., genetic algorithms) in Pyomo to optimize renewable energy systems.
- **Technical Toolset:** Python (PyTorch, Scikit-learn, TensorFlow, Pyomo, PyWake), MATLAB/Simulink, Modelica, QBlade, HOMER Pro, GIS, Aspen Plus, Typst, LaTeX.

### Honors & Awards

- **Erasmus Mundus scholarship** for DENSYS Master Program
- **JNED Award** for research on Japan's nuclear facilities (2023)

### Language

- Vietnamese (Native)
- English (IELTS 7.5 - Full Professional)
- German (A2), French (A1)

### Short Courses & Certifications

- **Self-Directed Learning - Energy Markets:** DTU course [Renewables in Electricity Markets \(46755\)](#) ([Personal Notes](#)). Topics: market structures, pricing mechanisms, day-ahead clearing, intraday trading, balancing markets, ancillary services, stochastic bidding.
- Wind Energy - Technical University of Denmark (DTU)
- 4th Training Course on Nuclear Power Plant Technology - JINED (studied in Vietnam and Japan)
- Optimisation with Python: Complete Pyomo with Bootcamp A-Z | Hydro, Wind & Solar Power - École Polytechnique de Paris

### Hobbies & Interests

🏹 **STRAVA** [Chương Đăng Tạ](#) | [Strava Profile](#)

Active Training

🏊 Swimming 🚴 Cycling 🏃 Running. Aspiring to complete a triathlon!

### Referees

#### Prof. Fabrice Lemoine

Chair of the DENSYS Erasmus Mundus Joint Master Degree, Université de Lorraine

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#### Prof. Marta Gandiglio

Associate Professor, Department of Energy (DENERG), Politecnico di Torino

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